THE COMPLETE GUIDE TO

FENCING

BARTH | BECK | JANKA (EDS.)

PREMIUM

MEYER & MEYER SPORT

The Complete Guide to Fencing

For reasons of readability we decided to use the male (neutral) speech form throughout this book, which of course also includes the female form.

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THE COMPLETE GUIDE TO

FENCING

BARTH | BECK | JANKA (EDS.)

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PREFACE

This book is an excellent example of an extremely successful symbiosis between theory and practice in a tradition-steeped and spectacular sport.

Emil Beck, Berndt Barth, and Claus Janka, three outstanding fencing coaches and instructors, brought together successful fencing coaches and scientists and published *Complete Guide to Fencing*.

Emil Beck, the brilliant coach who passed away in 2006, intuitively developed a completely new training method for fencing. He virtually embodied the athletic success of the German fencers. By now his methodology and the modern fencing style he shaped have significantly gained broad international recognition and emulation.

Berndt Barth, a longtime professor of training science at the prestigious German College of Physical Culture in Leipzig, Germany, has laid the scientific foundation for the training of several generations of certified physical education instructors and fencing specialists. His work at the DOSB's (German Olympic Sports Confederation) Coaches Academy in Cologne, Germany, and on the editorial board at the magazine *Leistungssport* (Competitive Sports) ensures him access to all the current findings in training science.

Claus Janka, graduate of the German College of Physical Culture in Leipzig, Germany, was a successful coach for former East Germany's Fencing Federation. Long before the fall of the Berlin wall, he had many dealings with Emil Beck, with whose explicit support he became the athletic director of the German Fencing Federation in 1990. His longtime work as an international official and functionary on the FIE's international executive committees are the basis for his complex views on the worldwide development of the sport.

The book *Fechttraining* (German) and *Complete Guide to Fencing* (English) contains all the important content of the training theory of fencing as well as tools for its practical implementation in the three fencing disciplines: foil, epée, and saber. Next to

the peculiarities of women's fencing and youth training, this book not only covers the psychological support and medical issues, but also suggests on how the sport of fencing can evolve and how it should be presented to the public.

This book, which has already been published in its second revised edition, marks the first time the entire body of knowledge on the sport of fencing, as it has been successfully practiced for years in Germany, has been contained in one work. All of the most significant international trends have been taken into account.

It was an excellent idea to also publish this book in the English language and to make it accessible, particularly to young up-and-coming fencing nations, and in doing so help them to rapidly reach the international fencing niveau and thereby continue to strengthen and promote the sport of fencing around the world. It creates a uniform basis for the modern training of coaches and greatly contributes to international communication. The fact that the first editions sold so quickly confirms that there is a ready market.

As director of the Coaches Academy of the German Olympic Sports Confederation in Cologne, Germany, and a member of the International Council for Coaching Excellence (ICCE) as well as the European Coaching Council (ECC), I wish the users of this reference book on fencing much success in their work for the good of the sport of fencing worldwide.

Prof. Dr. Lutz Nordmann
Director, DOSB Trainer Academy, Cologne, Germany

PREFACE BY THE AUTHORS

With *Tauberbischofsheimer Fencing Lessons*, Emil Beck, in 1978, published the foundation of his education and training system established in 1952.

One year later, under the direction of Berndt Barth, a group of authors consisting of trainers and training scientists from the German Fencing Confederation (DDR) published the second, much revised edition of their textbook *Fencing*, with the experiment of fencing training based on scientific theories.

It is an open question whether or not the decisive factor was the vision of Emil Beck or the reference by the Federal Commission for the Advancement of Competitive Sports to the yet to be created "training doctrine in the sport of fencing" listed in the preface to the *Tauberbischofsheimer Fencing Lessons*.

At any rate, Emil Beck used his first personal meeting with Berndt Barth on the occasion of an international épée tournament in the spring of 1981 to suggest collaboration on the publication of a book. "You have a great theory, and we are more successful in the practical area. That lends itself perfectly to a collaboration."

Right after the Olympic boycott in 1980, and with the particularly tense relationship between the two German nations, this was an almost utopian suggestion.

Eighteen years later, the time hat come. A team of twelve trainers, scientists, and journalists actively working in the sport of fencing implemented the plan—teaching training in the sport of fencing. In doing so, they succeeded in combining the different attempted training concepts of both former German fencing associations into one uniform and innovative training doctrine.

The authors chosen for this work represent the current German training doctrine in the sport of fencing. It clearly was the authors' intention to retain the proven contents and the overall structure of the forerunners; to integrate newer findings, theories, and methods from contributing exercise science-oriented fundamental sciences; to edit them for application in fencing training; and to supplement them with generalized teaching methods and experiences from successful trainers.

At the same time the manuscripts from those authors of the textbook *Fencing* from 1979, who are not involved in *Fencing Training*, were of some use. This applies in particular to Dr. R. Frester (psychology), Dr. K. Gottschalk (medicine), Dr. F.-J. Müller, (saber), P. Stanitzki (épée), and the illustrations of H. Hausmann.

The Complete Guide to Fencing is a reference book for training in competition-oriented fencing. Structure and content are geared toward the demands of coach and trainer instruction. It is a great source for suggestions to fencers who want to reflect on their training. Not everything in this book will be new to everyone. There also won't be a completely satisfactory answer to some questions. Sometimes they don't even exist yet. But readers should still have confidence in the book's statements. Much of it is theory driven, wherever possible backed by exercise science, and supported by the enduring success of German fencers[1] over the years.

The purpose was not to create a recipe book for fencing training, but rather to impart the knowledge needed to design one's own training. In the words of Hermann Hesse: "Practice should be the result of thinking, not the reverse".

We are always open to advice and suggestions.

The Authors

1 In view of the frequent use of the word "fencer" we would like to emphasize that these statements equally refer to female fencers. Special annotations are included in matters that are specific to women's fencing.

INTRODUCTION: ESSENTIAL INFORMATION REGARDING TERMINOLOGY

Fencing terminology combines technical sports terms, i.e. language specific to the training process, competition language, which is primarily based on French regulations, as well as science terminology from disciplines of sports science.

The technical language used in German fencing was in turn shaped by the most varied developmental influences, but in particular by the international fencing masters and trainers who have been working in Germany for centuries. By development we refer to the influence of the nations leading the world-fencing scene at any one time, mostly Italy and France, later Hungary and the Soviet states. The fencing masters that coined terminology came primarily from Italy, France, Hungary, Poland and Russia.

Here is a select example to chracterize the situation:

"Rimessa" (also "Rimesse") is the Italian term for an attack that continues, is repeated, or prolonged on the same, or—via disengagement—focused on a different target area. However, in French terminology one differentiates between "Remise" (same target area) and "Reprise" (different target area). If, in addition, any extra footwork is required, what the Italians will call a "Reprisa d'Attacco" (also Raddoppiamento dell'Affondo") becomes a „Reprise d'Attaque (also "Redoublement") for the French, while in the English-speaking world it is called "Continuation of attack". The Russians differentiate the "Powtornie Ataki" according to the opponent's behavior, depending on whether he parries, performs an incorrect riposte, or delays the riposte.

For instance, the French will occasionally also refer to a "redoublement" as only the footwork—lunge-dragging the supporting leg forward—as a lunge. In striving towards adequate German translations of these variations one struggled with continued attack, continuation of attack, and repeat attack, etc., without being able to vary the semantic content of these terms. Add to that translation errors and occasional misspellings in adopted terms, that were taken for granted and have become standard terms in some terminologies. "Cercle" is an example of this.

And then there are always the "fundamentalists" from a particular "school", who will only accept one view (sometimes the only one they know), or those whose views belong in historic essays and who cannot accept that the terminology evolves along with the fencing methods.

The following considerations serve as a **basis for communication** that is the foundation of this book:

Competitive fencing is a modern dueling sport characterized by the contest between two fencers. This occurs by means of planned performances and action sequences based on conscious decisions, which are called strategies. This is the case in military theory, psychology, mathematical game theory, politics, and, aside from some other areas, also in competitive sports. This is explained in Chapter 2.4. When fencers analyze a fencing competition with regard to strategy, the fencers' reciprocal actions can be identified as strategic elements according to their place and character. Thus strategic elements are actions that can, from a strategic aspect, be defined as attack or defense elements respectively. Their main and sub groups (basic strategic elements) are characterized by the same or similar strategic characteristics. That is why such a classification does not only contribute to communication between advocates of different schools, but also conveys overall conceptual clarity. In addition grouping, such as the system of basic strategic elements is based on, offer convenient starting points for the methodical design of training. This will become particularly apparent in chapters 6, 7 and eight.

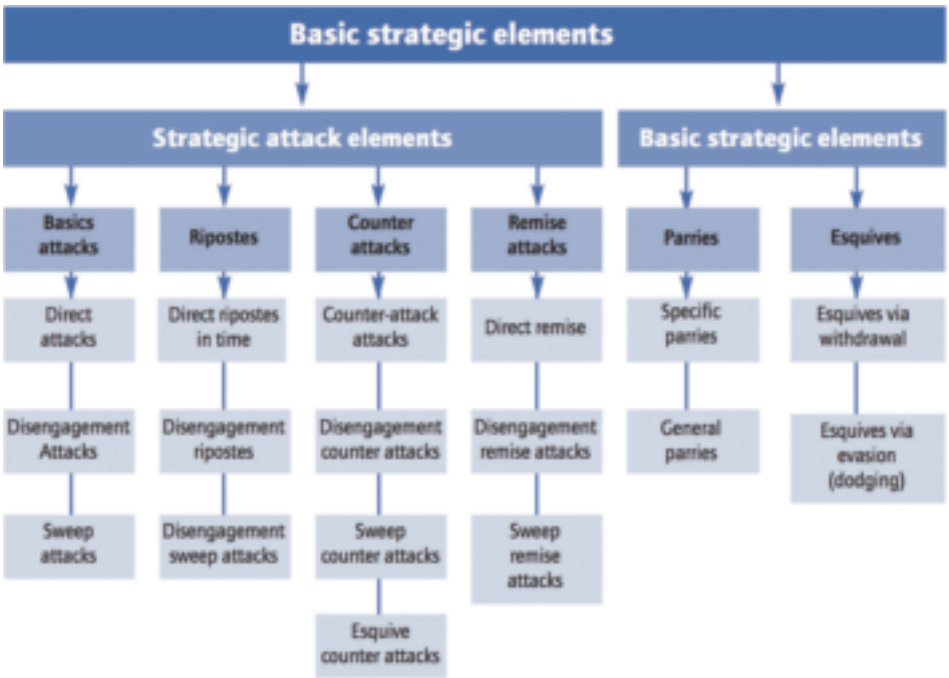

Fig. 0.-1: Classification of basic strategic elements

Strategic attack elements are attack actions that have the goal of making a hit.

- *Basic attacks* are attack actions that are executed against a non-attacking opponent without a direct link to previous attack or defensive actions.

Specific strategic elements, that are identified in the technical language according to the appropriate touch, disengagement or sweep actions and the position of the weapon, as direct thrust, direct cut, angulated thrust, disengagement thrust, cutover-cut, sixte-bind-thrust, quarte-beat-cut, sixte-glide, flick, can be found under the group or class name for basic attacks.

- *Ripostes* are attack actions that immediately follow a successful defensive action. They respond to an opposing attack action. One differentiates between direct riposte, disengagement-riposte, cutover-riposte, glide-riposte, etc.
- *Counter attacks* are attack actions that take direct advantage of the opponent's preparatory actions or direct attack actions for one's own attack. They are identified as disengagement in time, cutover in time, counter thrust, stop hit, block thrust, prise de fer, etc.
- *Remise attacks* are attack actions that follow directly after a failed attack. Like the other attacks, they are identified by the type of weapon movement in relation to the opposing blade, i.e. as direct, disengagement, and sweep-remise attacks.

Strategic defense elements are actions that have the goal of fending off the opponent's attack actions.

- *Parries* are defensive actions that prevent an opposing hit through movements with one's own weapon. They are referred to as septime pressure parry, octave beat parry, quart glide, sixte counter pressure parry, etc. One differentiates between general parries (defends multiple target areas and planes) and specific parries (most often in response to provoked or anticipated, therefore known attacks).
- *Esquives* (displacement parries) are defensive actions that prevent an opposing hit through body movements. They are referred to as retreating, dodging to the right, ducking, volte, etc.

For the sake of clarity, a conscious decision was made to forego listing each individual and possible strategic element. Every fencer can define his fencing actions if he refers to the main groups and the description of the fencing technique elements. Using the French and Italian terms shows that the terms of historic origin and those used in the FIE-Reglement can easily be incorporated. It is not a coining of "new terminology", but the creation of a systematic framework.

In practical fencing, established names were used for some sequences of basic strategic elements and for certain strategies. This applies in particular to "feint attacks, parry ripostes, and attacks with second intention."

- From a strategic-tactical point of view, feint attacks are combinations of two or more attack elements. The first attack (with double feint attacks the first and the second attack) serves to provoke mostly anticipated parries, to uncover unprotected openings, and to make a hit with a disengagement attack.
  All strategic attack elements serve as feints. The feints are identified by the strategic attack element that is used as a feint, by the number and type of parries, and by the attack element that finishes the feint attack.
  Example: battuta—thrust feint—disengagement thrust, feint-riposte with cutover, feint-counter attack (feint in time).
- *Parry-ripostes* are combinations of parries and ripostes. The favorable measure created by the opposing attack is used to make a hit. They are identified by the parry that was used and the riposte that was used.
  Example: Septime press parry—disengagement riposte, quart glide-(direct) riposte, sixte parry-glide riposte, etc. A parry riposte that follows a parry-riposte is called a counter parry-riposte.
- *Attacks with second intention* are combinations of two attack elements and one defensive element, or of two attack elements. With a feigned attack the opponent's attention is diverted from the true (second) intention and he is challenged to a parry-riposte or a counter-attack. The hit is made with a second parry-riposte (counter parry-riposte) or a counter attack (counter time).

The classification of technical elements that are the basic components of every movement action in fencing is done in a similar way. The structure of movements and their function are the classification characteristics. This makes it possible to describe the special characteristics of each individual group, as well as the positions and movements, in a generally binding and non-weapon specific way, to point out possible variations, and to strive for conceptual clarity.

We will forgo the classification and description of technical elements (basic technical elements). Basic references are provided in Chapter 2.3. The combinations are described in connection with the weapon-specific solution to the strategic-tactical tasks in Chapters 6, 7 and 8.

CHAPTER 1

1 DIVIDED LIKE THE COUNTRY—THE SPORT OF FENCING IN POST-1945 GERMANY

2 PERFORMANCE STRUCTURE, PERFORMANCE CAPACITY, AND PERFORMANCE DEVELOPMENT

3 BASIC METHODS OF FENCING TRAINING

4 YOUTH TRAINING

5 ASPECTS OF SPORTS MEDICINE IN FENCING

6 FOIL FENCING

7 EPÉE FENCING

8 SABER FENCING

9 INTERNATIONAL DEVELOPMENT TRENDS

10 PRESENTATION OF THE SPORT OF FENCING

APPENDIX

CHAPTER 1

1 DIVIDED LIKE THE COUNTRY—THE SPORT OF FENCING IN POST-1945 GERMANY

Like the country itself, the history of German fencing after 1945 was divided until the fall of the Berlin Wall. As Europe lay in ruins after World War II, there could be no immediate thought of fighting with épée, foil and saber. The Allied Forces had made sure of that on Dec. 17, 1945, with the Allied Control Council Directive 23. They viewed fencing, as well as any other martial arts sport, as a paramilitary exercise and a threat to peace. It took four years until the Allied forces revoked the ban and realized that the fencers with their blunted weapons were no bloodthirsty warriors.

Meanwhile, the Iron Curtain had been closed, East and West were no longer just cardinal directions, and the Federal Republic of Germany and the German Democratic Republic were a political reality. At first, enthusiasts in the West made efforts toward the **renewed founding of the German Fencing Federation (DFB), which was launched on Dec. 17, 1911, in Frankfurt**, on the initiative of Jacob Erckrath de Bary. Arbitrarily, meaning without the authority of the Allied forces, they tackled the reconstruction of the DFB. A foundation assembly was held on November 27, 1949, on the floor of the upper house of the German parliament in Bonn, which was attended by representatives from 157 fencing clubs and departments. Only six months later, did the Allied High Commission indirectly approve this surge ahead, by officially abrogating the ban.

In the DDR (German Democratic Republic), it took a few more years for the reconstruction of the sport of fencing, which was viewed as elitist, to begin. **On June 15, 1952, the**

"Fencing Section" was founded at the sports commission in Berlin, where Karl Fischer, Herbert Schmiedel and Albert Gipp successively took charge. In addition fencing departments were formed in Leipzig, Erfurt, Magdeburg, Halle, Berlin, Zella-Mehlis, Gera, Rostock, Chemnitz and Dresden. At the end of 1951, 47 sections had already begun their training operation.

The first fencing tournaments within the DDR were already being held in Leipzig in July and October of 1951. The first DDR championships were held a year later, where Georg Neuber of Motor Dresden became champion in both foil and saber. But there was a shortage in fencing attire, and especially in blades, which led to limited training and competition activity.

Nevertheless, it became possible to expand the activities to an international level. At the end of 1952, Czech and East German fencers competed in Prague and Gottwaldov. This was the prelude to numerous competitions with other socialist countries, of which the USSR and Hungary, with their know-how, provided a significant impetus for increased performance.

In spite of intensive work in the youth area, fencing initially remained a sport that inspired little interest. By 1961, the number of organized members of clubs and departments had barely risen to 4200; only after 1971 did that number go up to about 7500. Not enough to produce top fencers that could bring international glory and success.

THE KIDNAPPING OF THE FIE PRESIDENT AND ITS CONSEQUENCES

But before German fencers were once more permitted to compete on the world scene, feelings of resentment within the international fencing organization FIE had to be aired out. Although the FIE president at the time, Jacques Coutrot, whose godfathers included the German Erckrath de Bary, on Nov. 29th, 1913, pleaded for the reentry of the DFB, it was still rejected by the FIE Congress in 1951.

It had not been forgotten that the FIE president Paus Anspach, who was in office from 1933 to 1948, had been kidnapped in the summer of 1940, by Reinhard Heydrich, the head of the Gestapo. The reason: fencing enthusiast Heydrich himself wanted to become FIE president. To this purpose he kidnapped Anspach and had the entire FIE archive confiscated and brought to Berlin, where it burned in 1945. The Belgian was also carried off to the capital, where he was aggressively, but unsuccessfully, asked to resign.

This knavery bestowed upon the DFB another year of international ostracism. Only at the FIE Congress on March 28, 1952, did a majority vote for the reentry of the (West-)

Germans. This meant that the DFB fencers' participation at the Helsinki Olympics, which marked the beginning of a slow but glorious climb to the world's fencing elite, was secured.

The fencers in East Germany had to wait longer to become members in the world association. After two applications for membership, in 1952 and 1954, were declined, the 37th FIE Congress decided on May 12, 1956, in Milan, to admit the Fencing Section of East Germany. Thus the road to the international title competitions was now open to the Germans from the East as well.

An organizational prerequisite to this was the founding of the German Fencing Confederation (DFV) on July 20, 1958, at the central delegates' conference of the Fencing Section in Leipzig. Alfred Röll became the first president, followed by Rudolf Hansen and finally Rolf Borrmann. The latter led the organization for 20 years, until 1990.

The first big showdown between the now internationally acceptable fencers from East and West came before the Olympic Games in Rome in 1960, and in Tokyo in 1964, where united German teams participated for the last time. During the elimination rounds the East German fencers suffered great disappointments, as none of them could prevail against the rivals from the West. This was particularly bitter before the Japan Games, because the good results at the World Championships in Danzig in 1963 had inspired much hope. The defeat of the repeated team-world champions Italy by the East German épée fencers, and the consequent fifth place finish, was the first major international success for the DFV.

But East Germany's fencing revival was limited. The small base of talented junior fencers was one reason, the lack of broad promotion of this cost-intensive individual sport, was the other one. Fencing, where international contacts and the global competition advance success within its community, did not fit into East Germany's competitive sports system. Anyone who could not guarantee medals found little support.

For the active fencers in the DFB however, the whole world was wide open, yet successes did not come automatically and in huge numbers. During the 50s and 60s, big celebrations were only sporadic. Ilse Keydel, who got third place in Brussels in 1953, won the first World Championship medal after World War II. But one name stood out above the rest: Heidi Schmid. She took second place at the World Championships in Paris in 1957, at the tender age of 17, exactly 20 years after the final World Championship victory of the legendary Helene Mayer, thus steeling the female German foil fencers.

The fencer from Augsburg created a sensation at the Rome Olympics, where she won a gold medal on September 1, 1960. In the following year, she also won the World

Championships in Torino, Italy, as well as the students' World Championships, and was the first female fencer to be voted "Athlete of the Year". The thread of success broke in Tokyo, host of the 1964 Olympics: Heidi Schmid was eliminated in the intermediate round. But Helga Mees of Saarbrücken made the breakthrough and won the silver.

During that time, which was shaped by the formative influence of fencing geniuses such as Christian d'Oriola (France) and Eduardo Mangiarotti (Italy), the male German foil fencers did not get a stab in. However, as a team they made up an accurate group, which won World Championship-silver behind the upcoming fencing power, USSR, in Budapest 1959. One year later, at the Olympics, they also landed the bronze medal.

But it took nearly another decade before the foil of an exceptional German fencer could keep the world elite in-check: Friedrich Wessel. The Bonn fencer immediately landed a double coup by securing the World Championship title both in Cuba in 1969, and in Ankara in 1970. The Olympic crowning of his career was denied this pure amateur, who passed up participating in the 1971 World Championships in favor of pursuing his law degree, which likely cost him a medal one year later in Munich.

At that time, gold, silver, and bronze medals in épée and saber were only the stuff dreams were made of. Only in the era of Emil Beck, the originator of the (West-) German fencing miracle, would the entire armory get a new polish, whereby the Germans only broke through the phalanx of the Eastern Europeans in saber after its electrification in the mid-80s, combined with new training methods and rule changes.

FENCING DID NOT FIT INTO THE EAST GERMAN SYSTEM

East Germany lacked the resources, the freedom of movement, and the support for someone like Beck to create a similar revival of the fencing sport, such as was possible in the other Germany. In view of the sparse sponsorship by the state, for which medals and world records were symbols of the socialist system's superiority, it is in fact surprising that East Germany was able to claim any successes at all. Initially the saber was the hardest-hitting weapon there. At the Olympic games in Mexico City in 1968, the East German team took a remarkable fifth place, and was fourth three years later, at the World Championships in Vienna. But the East German fencers could only come close to medaling, such as Torsten Kühnemund of Potsdam, whose fifth place finish at the 1988 Olympics represented the best result in international individual épée fencing.

The big scores in foil fencing started to happen after Claus Janka took charge of that weapon within the East German organization. At the World Championships in Hamburg

in 1978, and at the Boycott Games in Moscow in 1980, the East German male and female national foil fencing teams moved into focus with a fourth place finish respectively. Incidentally, in the year between, a female quartet that included Mandy Niklaus, had won gold at the student World Championships. In 1982 in Rome, the athlete from Dresden United won the first (bronze) medal for East Germany at fencing World Championships. Of all things, a West German fencer, Sabine Bischoff, nearly bungled that historic success. She sent Mandy Niklaus into the semi-finals, but in the final round she got revenge in the fight for bronze.

Peter Proske was one of her trainers, whose subsequent protégé, Udo Wagner of Dresden, won Olympic silver in foil fencing in 1988, thus presenting East Germany with its greatest fencing triumph to date. Like many of his colleagues, Peter Proske, who was the athletic director of the Tauberbischofsheim fencing club until 2004, after the turning point, switched over to training centers throughout Germany.

The world's most successful fencing club in Tauberbischofsheim also became a magnet for a number of top athletes from East Germany. After the fall of the Berlin wall, Uwe Römer, Ingo Weissenborn, and Udo Wagner moved to the Baden region and with their skills were able to prevail even in the "new Germany".

The last East German champion, Weissenborn, became the first foil fencing world champion of the reunited Germany in Budapest in 1991, namely in the duel for gold against Thorsten Weidner (TBB). "After this victory I feel liberated", Thorsten said at that time. He had received his first lessons from Lok Bernburg, and had drawn attention with his spartakiad victories in foil fencing and saber. Together with his companion Wagner, he was part of the East German foil fencing team that won World Championship silver in 1983, and bronze in 1986, as well as the German team that won the Olympic title in 1992.

SYNONYMOUS WITH FENCING: EMIL BECK

In the old Federal Republic one man, whose name would become synonymous with fencing, set new standards: Emil Beck. Born on July 20, 1935, in Tauberbischofsheim as the youngest of a family of 13, and first becoming aware of the sport in 1951 at a movie theater, he became the epitome of the medal smith. He created a new fencing school, built an efficient training center around the "FC TBB", cleared many obstacles out of the way, and by the early 70s had made the small town in the Baden region world famous.

The statistics of Beck's success are phenomenal. During the three decades from 1968 until 1998, his Tauberbischofsheim fencers won tons of titles and medals: 18 Olympic medals,

74 world and 28 European championships are a proud record. Add to that 34 European and 129 World Cup victories, as well as 399 German national titles. This achievement is unparalleled in the world and is essentially credited to Emil Beck, who gave up direction of the Tauberbischofsheim Olympic base on July 31, 2001, ending his career.

Of all things two triumphs in épée fencing, the most fickle of all weapons, marked the starting point of a new fencing era. In 1973, the German team won a sensational gold medal at the World Championships, providing the "Goeteborg Miracle". At the same time Beck experienced the first low point of his career at the Olympics in Munich in 1972, when the German fencers, against all expectations, did not win any medals.

That was definitely not the case in Montreal in 1976. Alexander Pusch, the épée genius, refined his 1975 World Championship success with an Olympic victory, which he snatched away from his teammate Jürgen Hehn. In the end the club also won silver. The big Olympic coup was landed by the men's foil fencing team, of which Thomas Bach was a member. He later continued the parry-riposte game in the sports-political arena on the International Olympic Committee (IOC), where he moved all the way up to vice president.

Since the Montreal Olympics, Germany ranks amongst the fencing elite such as Italy, France, or Russia. Since then there have been many highlights and a few lows, as well as a number of excellent athletes whose talent and character have given a face to this fringe sport.

Alex Pusch, who actually won his third title at the 1978 World Championships in Hamburg, was a kind of "Franz Beckenbauer" of fencing. Amongst the community of épée "players" in the 20th century, this was only accomplished by Italy's miracle weapon, Edoardo Mangiarotti, and Alexander Nikantschikow from the former USSR, as well as the Frenchman Eric Srecki. When Pusch, spoiled by his early successes, did no longer produce, the star of Elmar Borrmann, who in 1983 became the second German épée world champion, appeared on the horizon. Seven years later, Thomas Gerull surprisingly managed to do the same. Four years later Volker Fischer did it.

But after that the stage belonged to Arnd Schmitt of Heidenheim, who, like Pusch, was gifted and headstrong. Like his famous predecessor, he also became Olympic Champion in Seoul in 1988, but unsuccessfully struggled to become individual world champion at ten differernt World Championships (1985-1998). Only in 1999, was he able to break the spell and,—once again in Seoul—won the world champion's crown he was missing. Oliver Lücke, who only won his first (bronze) medal at the World Championships in 2001 at age 37, had to wait a long time for a big individual success.

South Korea was not only a highlight in Schmitt's career, but also a general milestone in the history of the DFB. This was largely due to the female foil fencers, Anja Fichtel, Sabine Bau and Zita Funkenhauser, who, in that order, won gold, silver and bronze, and together won another gold medal. This women's trio was the most effective team German fencing has ever produced. In the forefront was Anja Fichtel, whose successes and charisma overshadowed Helene Mayer, Heidi Schmid and Cornelia Hanisch. But "Conny" Hanisch of Offenbach, who produced a World Championship title-trilogy with her wins in 1979, 1981 and 1985, also reached enormous popularity, which even allowed her to win the title of "female athlete of the year" (1986).

Anja Fichtel in fact never received this honorary title, but with her Olympic triumph, her two individual World Championship gold medals (1986/1990), and a total of ten World Championship medals, as well as ten national individual titles, she outstripped all of her predecessors.

Moreover, this Tauberbischofsheim fencer became the darling of the media like no other fencer before her. Much to Emil Beck's displeasure, she was quite outspoken, going on the attack off the strip, posed in the nude for magazines, and did not let pregnancy and the birth of a child interfere with her career. Sabine Bau, who landed a big surprise of her own after her superior rival retired, did not manage herself as spectacularly. In 1998 she finally got lucky and triumphed at the World Championships in La Chaux-de-Fonds. Twice more, in 1999 and 2001, she managed to land World Championship silver before she ended her career in 2003. At the Sidney Olympics in 2000, Bau had to leave the limelight to her Tauberbischofsheim teammate, Rita König, who got second place.

Many say that becoming the best in the world in men's foil fencing is fencing's crowning glory. Five German athletes have accomplished this: Friedrich Wessel was the first in 1969 and 1970, then Matthias Gey in 1987, Alexander Koch in 1989 and 1993, Ingo Weissenborn in 1991, and Peter Joppich in 2003. Only one touch kept Ralf Bissdorf from becoming the first German Olympic champion in foil fencing at the Olympic finals in Sydney in 2000. Duplicity of events: at the Olympic games in Los Angeles in 1984, Matthias Behr also missed the gold medal by one touch.

Only one German could become number one in saber fencing at World Championships in the 20th century, namely Felix Becker, who was world champion in Athens in 1994. One of the precursors preparing the laborious ascent into the international elite was Jürgen Nolte, who won eight national championships between 1980 and 1990, and was a noteworthy sixth at the 1986 World Championships; better results had been achieved before him by Gustav Casmir with a medal win at the (unofficial) Olympic in-between

games in 1906, and his nephew Erwin Casmir, who achieved even better results with a fourth place fnish at the 1936 Olympic Games Berlin. In the late 90's, a new generation of German saber fencers prepared to set out. At the forefront was Wiradech Kothny, who had become European champion in 1999, and had won two bronze medals (individual and team) at the Olympics in Sidney in 2000.

Women's saber fencing only gained admittance to the World Championships one year before the turn of the millennium and has been part of the Olympic program since 2004. Of the German lady fencers Sandra Benad became a successful pioneer in this new discipline. The Eislingen native was able to take third place in the 2000 World Championships. One year later, at the European Championships in Koblenz, Germany, she won individual silver and led the saber team to win the title.

The women have been competing in épée fencing at the World Championships since 1989, whereby the Germans have been amongst the world elite since the beginning, as the team victories from 1988 (at the yet unofficial World Championships) and 1990, or the second place finish at the 2003 World Championships demonstrate. The DFB had to be patient until 2001, when Claudia Bokel was able to pocket the fist individual World Championship title. But other experts, such as Ute Schaeper, Eva-Maria Ittner, Katja Nass, Denis Holzkamp, and Imke Duplitzer, were also getting close to reaching this giant goal. They were runners-up at the World Championships in 1989, 1991, 1994, 1998 and 2002, and won the silver medal at the Olympic games in 2004.

But the success of an athletic association does not only consist of athletic merits. One of the greatest challenges for German fencing was the German reunification and the coalescence of the two associations. The fact that the sports-political threads were spun with the lighter foil, abandoning the rattling of sabers, was proven at the German Fencing Congress on December 8, 1990, in Bonn. By the sportsmanlike professional interaction between the fencers and the preparedness of the majority of representatives of both German fencing associations even in times of political confrontation, the regional associations Mecklenburg-Vorpommern, Brandenburg, Sachsen, Sachsen-Anhalt, and Thuringia were admitted to the DFB in the spirit of chivalry, and the seal was set to the union of fencers.

"This is our contribution to fair play. For us that term is not just lip service", said DFB president, Erika Dienstl. In 1986, the native of Stolberg near Aachen in the Rhineland, was the first woman after Erwin Casmir (1949-1957), Otto Adam (1957-1972, Elmar Waterloh (1972-1978), and Klaus Dieter Güse (1978-1986), to be elected to the top of the DFB.

Fig. 1-1: The "unification presidents": Erika Dienstl for the DFB (West) and Dr. Berndt Barth for the DFV (East) with the 1990 unification motto.

The final act of the DFV of the former East Germany was its disbandment on Dec. 31, 1990. In May of that year, Dr. Berndt Barth was elected president of the DFV. The last chairman of the East German organization became vice president, and later secretary general of the DFB. The fact that another man from the German East, Claus Janka, also became sports director proves, that the fencers wanted unification also for the sake of fair play.

It was not expected that fencing, like some other sports, would see a long-term performance increase due to reunification. A number of excellent athletes from the new German states were able to make the leap to the national teams, but the continuous, systematic, and successful schooling of talent for world-class competitions developed only slowly. However, there is a lively fencing culture in cities like Berlin, Potsdam, Leipzig, Dresden, Halle, or Jena that may become the basis for a new revival in the East.

Fig. 1-2: Peter Joppich (right), four-time world champion in men's foil individual.

But in the 2000s, an athlete from Western Germany ascended to become one of the best fencers in history: Peter Joppich. The foil specialist who was born in Koblenz, Germany, and coached by Ulrich Schreck in Bonn, Germany, became world champion for the first time in 2003 and went on to win three more world championship titles in 2006, 2007, and 2010.

Only the Frenchman Christian D'Oriola has won more world championship titles.

On March 12, 2006, sad news reached the fencing world. Emil Beck, the great coach and reformer from Tauberbischofsheim, Germany, had passed away.

At the start of the new millennium, Britta Heidemann, Benjamin Kleibrink, and Nikolas Limbach advanced to join Joppich to become the "faces of German fencing". Not only did Heidemann become world champion in 2007, but one year later, at the Peking Olympics, she and Kleibrink created a moment of glory for the DFB (German Fencing Federation), the likes of which had not been seen in Seoul since 1988. Within minutes both won Olympic gold.

Fig. 1-3: Britta Heidemann (right), Olympic, world, and European champion in women's epee individual.

The 2009 World Championships represented a milestone for the DFB. Nikolas Limbach became the second saber world champion after Felix Becker (1994). He is the outstanding athlete at the saber school built in Dormagen, Germany, by national coach Vilmos Szabo that becomes an international trademark. However, Limbach had to wait until 2014 to become world champion for the first time with the German saber team. This was followed by a European title win in 2015.

The DFB, which celebrated its centennial on December 17, 2011, achieved a success in sports diplomacy at the FIE's world federation congress. Germany won the bid for hosting the 2017 World Championships in Leipzig, Germany.

CHRONOLOGY OF THE HISTORY OF GEMAN FENCING POST-1945

(Compiled by Andreas Schirmer)

12.17.11	Founding of the DFB in Frankfurt
1939-45	World War II
12.17.45	Allied Control Council Directive 23 for the ban on fencing in Germany.
West Germany	
11.27.49	Re-founding of the German Fencing Federation (DFB) in Bonn
1951	First post-war German championships.
3.28.52	Readmission of the DFB into the World Federation F.I.E.
1953	Ilse Keydel wins the first World Championship medal.
9.1.60	Heidi Schmid becomes Olympic champion
1969	Friedrich Wessel becomes first German foil champion.
1973	First World Championship victory by German épée team.
1975	Alexander Pusch becomes first German épée world champion.
1976	Pusch and DFB men's foil team become Olympic champions.
1978	First World Championships in Germany (Hamburg) – Pusch world champion.
1979	Cornelia Hanisch wins first of three World Championship titles.
1980	Boycott of Moscow Olympics.
1984	German épée and women's foil team Olympic champions.
1986	Anja Fichtel becomes youngest world champion in history.
1988	Seoul Olympics. DFB wins seven medals and nationas ranking, Fichtel, Schmitt, and women's foil team win gold.
1989	First time German World Championship medals in
	saber team (silver) and Felix Becker (bronze).
East Germany	
6.15.51	Fencing Section is founded by the German Sports Committee in Berlin.
5.12.56	East Germany's Fencing Section is admitted to the F.I.E.
7.20.58	Founding of the East German Fencing Association in Leipzig.
1963	East Germany's épée team gets fifth place at the World Championships in Danzig.
1968	East Germany's épée team gets fith place at the Mexico City Olympics.
1978	Women's and men's foil teams get fourth place at World Championships.
1979	Women's foil wins Students' World Championships.
1980	Women's and men's foil teams get fourth place at Olympics.
1982	Mandy Niklaus is first female East German fencer to win World Championship medal (3)
1988	Udo Wagner is second to win first Olympic medal in individual foil.
1989 German reunification	
12.8.90	Admission of regional associations "East" into the DFB.
1991	Ingo Weißenborn becomes first (foil) world champion after merging of fencing associations.

1992	German men's foil and women's épée teams become world champions for the second time.
1993	World Championships in Essen; hosts win 11 medals and nation ranking; Alexander Koch, men's foil and women's foil team world champions.
1994	Felix Becker becomes fist German world champion in saber.
1995	Atlanta Olympics; Arnd Schmitt flag bearer for entire German team; only women's foil team takes bronze.
1998	Exhibition competition in women's saber at World Championships; Sandra Benath takes seventh place.
1999	World Championships in Seoul; 6 medals; gold for Arnd Schmitt and women's foil team; Wiradech Kothny becomes European champion.
2000	Kothny wins first Olympian saber medal since 1906; five medals total in Sydney.
2001	First European Championships in Germany; Men's foil and women's saber teams win titles in Koblenz; Claudia Bokel becomes world champion in women's épée.
2002	At World Championships Andre Wessels takes second in individual and wins gold with men's foil team; Imke Duplitzer and Britta Heidemann take silver and bronze in épée.
2003	Peter Joppich becomes world champion; Simon Senft becomes European champion in men's foil.
2004	Women's épée teams won silver at the Athens Olympics; the men's épée team won bronze. Richard breutner becomes European Champion in men's foil.
2005	For the third time after Hamburg in 1978, and Essen in 1993, World Championships take place in Germany, this time in Leipzig.
2006	The great Tauberbischofsheim fencing coach Emil Beck dies. Peter Joppich becomes world champion in foil for the second time. Only one German fencer was able to do so before him: Friedrich Wessel.
2007	Peter Joppich wins his third world championship title; Britta Heidemann becomes women's saber world champion.
2008	Britta Heidemann and Benjamin Kleibrink win gold at the Peking Olympics.
2009	Nikolas Limbach becomes world champion as only the second German saber fencer after Felix Becker (1994).
2010	Peter Joppich celebrates his fourth world championship title. Only the Frenchman Christian D'Oriola won more foil world championships—namely, six.
2011	On December 17, the German Fencing Federation celebrates its centennial.
2012	Britta Heidemann wins silver at the London Olympics. Unforgotten is the dramatic semi-final against Shin A-Lam of China, which Heidemann won at the last second.
2014	The German Fencing Federation is chosen as host of the 2017 World Championships in Leipzig, Germany, at the FIE congress. A German saber team wins the world championship title for the first time at the world championships in Kazan, Russia.
2015	For the first time in history the German saber team wins the European Championships.

CHAPTER 2

1 DIVIDED LIKE THE COUNTRY—THE SPORT OF FENCING IN POST-1945 GERMANY

2 PERFORMANCE STRUCTURE, PERFORMANCE CAPACITY, AND PERFORMANCE DEVELOPMENT

3 BASIC METHODS OF FENCING TRAINING

4 YOUTH TRAINING

5 ASPECTS OF SPORTS MEDICINE IN FENCING

6 FOIL FENCING

7 EPÉE FENCING

8 SABER FENCING

9 INTERNATIONAL DEVELOPMENT TRENDS

10 PRESENTATION OF THE SPORT OF FENCING

APPENDIX

CHAPTER 2

2 PERFORMANCE STRUCTURE, PERFORMANCE CAPACITY, AND PERFORMANCE DEVELOPMENT

2.1 DEMAND AND PERFORMANCE STRUCTURE

2.1.1 DEMAND PROFILE AND PERFORMANCE STRUCTURE

Many fencing trainers picture the "ideal" fencer as "fast as a sprinter, a technical virtuoso like a violinist, and savvy as a chess player". Further attributes are elegant and fierce, goal oriented and persistent, strong willed and disciplined, and tall and slender respectively.

On the one hand this is correct because there are many individual factors involved in fencing performance. Fencing is certainly not comparable to sports in which the result or performance capacity is primarily defined by the pre-eminent significance of one criterion, like a marathon runner's endurance, a weight lifter's strength, or a gymnast's technical acrobatic skills. Even the most brilliant technician won't be successful in fencing if his action is not executed quickly and at the right moment, and the most persistent fencer will lose if he does not hit the opponent, but rather misses him.

On the other hand, fencers with good competition results have a rather varying appearance. There are world champions who are over six feet tall, and there are others who are around five feet tall. There are very elegant fencers, but also successfully fencing

athletes with an unorthodox fencing style. There are the savvy strategists and clever tacticians, as well as the relatively one-sided, very fast reacting and strong fencers. Most often, particular talents or developed strengths are the basis for the fencers' various, individually preferred ways of fighting (performance style). There is a strongly generalized differentiation between the so-called "fighters" and the "tacticians".

Fencers who are labeled *fighters* are those who can, due to good special-temporal orientation, act and react quickly, consequently and strongly—regulated by impulse—in quickly evolving fight situations.

Tacticians are fencers who anticipate emerging situations or actively prepare for them with tactical maneuvers respectively, and successfully handle the contest with the opponent based on predominately reflexive decisions after strategic-tactical considerations, aided by complicated motion sequences.

With this two-sided examination it can be concluded, that fencing is a very complex sport that requires a multitude of special performance abilities and qualities for successful execution. But it can also be concluded that these performance requisites are related and in part alternately complement or compensate (balance) each other respectively.

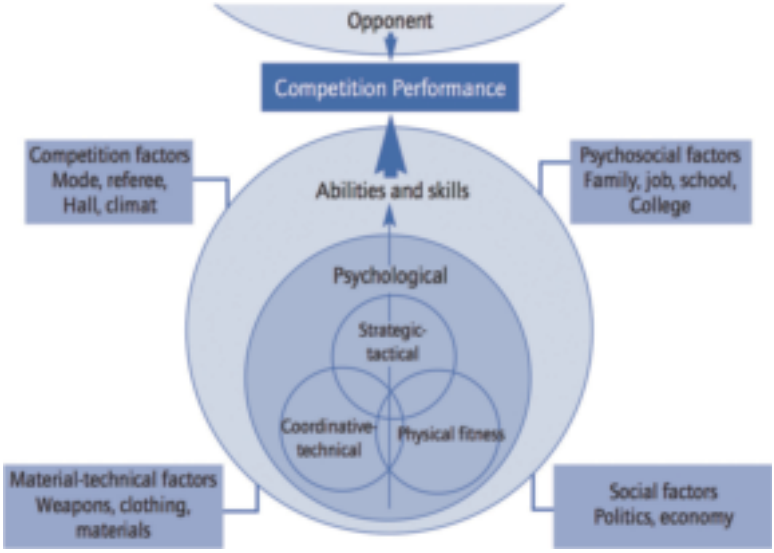

Fig. 2.1.-1: Schematic performance structure model in fencing

Previous opinions on the performance structure of fencing—the inner structure of the concrete fencing performance or the factors involved in achieving performance and their interdependence respectively—are based on requirements analysis and the idea that the

individual abilities and physical characteristics involved in the complex performance, as well as other components, can be combined into "complex performance factors". These complex factors are shown in schematically simplified form as a performance structure model.

At the core of the illustration are the *abilities and skills* that are tied to the *character system*. In the above illustration these are shown as circles:

- **Physical fitness abilities and skills** combined with the physical-organic prerequisites (build, support and musculoskeletal system, cardiovascular system, muscles, nerves, sensory organs, etc., and the respective developmental status of strength, speed and endurance).
- **Coordinative-technical (movement regulating)** abilities (muscular coordination, movement perception, movement visualization, ability to control movements) and developed **sports-technical skills**.
- **Strategic-tactical abilities, action-regulating skills and processes** (feeling, perceiving, visualizing, thinking), developed or acquired, respectively, strategic-tactical knowledge and ability, as well as the ability to act.
- Around this core the **psychological abilities and skills** form a relatively independent complex, because the psychological influence factors of cognition (planning, evaluating, decision making), incentive (interests, goals, desires, expectations) and emotion (joy, anger, confidence, insecurity) influence the three other areas.

External factors (schematically shown as rectangles) influence the fencer's specific ability to act (the performance and the result) insofar as the

- **psychosocial factors** (family, school, college, job, friends, etc.),
- **social factors** (environment, evaluation and endorsing of performance through social classification, public effects, allocating of resources, sports scientific findings, etc)
- **material-technical factors** (competition material, weapons, clothing, to the point of special sports equipment for training) and
- **competition factors** (competition mode, location of competition, performance evaluation by referees, etc)

can become performance benefiting or performance inhibiting. By controlling this entire complex Emil Beck organized and secured the successes of his Tauberbischofsheim fencers for many years.

The intertwining of the factors in the schematic illustration of figure 2.1.-1, suggests the reciprocal linkage. The comprehensive classification of the psychological abilities and skills symbolizes their general significance.

Modern future-oriented approaches in the examination of the performance structure try to illustrate the correlation between the factors, their "internal functioning", with an integral concept. It is that mutual conditionality or structural and functional linkage of biological processes of the energy regulation (fitness level), the sensorimotor processes of the movement regulation (technique/coordination), and the mental-cognitive processes of the action and behavior regulation (strategy/technique), combined with the situation-outlasting, character-specific (cognitive, emotional and motivational) dispositions, that characterize the substance of a fencer's performance and performance capacity structure.

For a better understanding of the suggestions for modern fencing training in the following chapters, they are preceded by a summary and brief explanation of the most important starting positions.

1. The athletic performance of a fencer is the complex result of the actions of a personality. Based on the performance capacity of the apparatuses and functions, the multifaceted linkages between abilities, characteristics and skills have an effect on the level of competitive performance, the attitude towards coping with the demand, and the individual ability to act.
2. The individual performance requisites (abilities, skills, characteristics, etc.) are, with regard to their completeness and their participation in achieving the athletic performance, not definable at present. Therefore fitness reports and exact performance predictions are only possible by approximation. The "bio-psychosocial entitiy fencer" manages the type of sport as well as the discipline-specific bias of the performance capacity (structurally and functionally), in terms of a "self-organization" through the frequent accomplishing—on training methodical grounds—of the specific demands in training and competition of the sport and the discipline. This process takes place irrespective of what we already know about it in detail.
3. There is no special ability "as such", but just the abilities of the fencer for concrete activities, actions or movements. That is why no claim of validity for other spheres is being sought with those statements regarding "personality" and its structure. Intended are references to the entire personality, with regard to the activity (actions, movements), but only to the specific areas of demand affected by the sport of fencing. A fencer's performance capacity is purposive. Fencers are highly specialized and therefore most often only successful in one discipline.

4. Since fencers accomplish equal athletic performances (procure victories) with very different physiques, different action repertoires, and different fighting manners, it is beneficial and essential to address the fencers' individual characteristics and to assess or to create respectively, compensation alternatives.

Typical compensation alternatives for fencers are compiled in table 2.1.-1, as per research results by Rodionov, 1979

Table 2.1.-1: Typical compensation alternatives in fencing (as per Rodionov, 1979)

Deficiencies	Compensation
In strategic thinking	Through speed of motor reaction and execution.
In the spreading of attention.	Through speed of perception and thought operations. Accuracy of muscular movement differentiation.
In the shifting of attention.	Through fast motor reaction, situation anticipation and feel for tempo.
In the speed of motor reaction.	Through feel for measure and tempo, situation anticipation, spreading of steady attention, strategic-tactical thinking.
In the accuracy of muscular movement differentiation.	Through constant attention, speed of motor reaction, good feel for tempo.

5. The fencers' individual performance requisites are structured "hierarchically" on different regulation levels, according to the concrete demands. That means, that the complex performance factors identified in fig. 2.1.–1, do not act side by side as a "performance structure" but are, as function- (also regulation) levels, inferior or superior in a hierarchical sense, are in a "double hierarchical relationship" to each other. With this type of "holistic concept" following synergistic conclusions (Pickenhain, 1996; Janssen et. Al, 1996) an attempt is made—admittedly with a much-simplified model—to explain the interrelationship between the sub-systems and their "internal functioning".

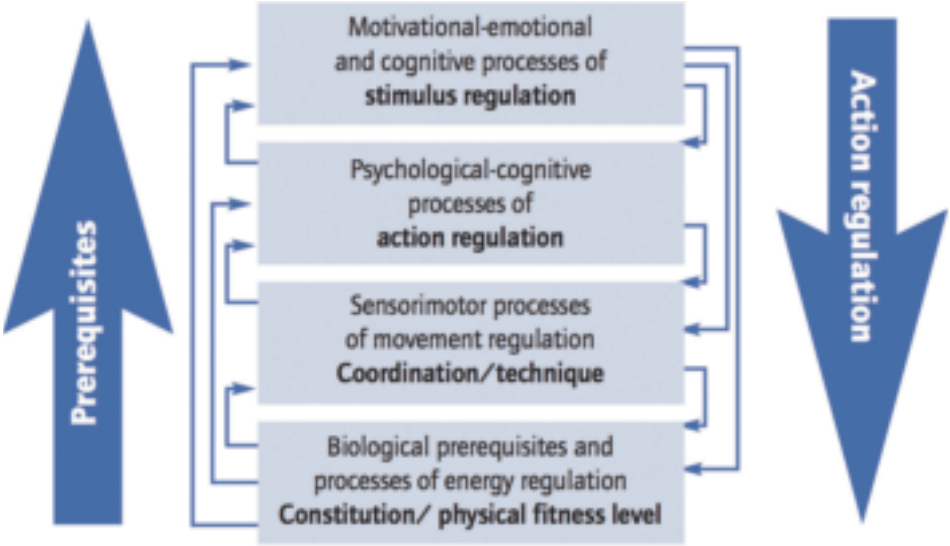

Fig. 2.1.-2: Relationships of function levels of the action system (schematic)

This "double hierarchy" becomes apparent in that, on the one hand, the organism's existence and the biological processes of the energy regulation in the, for the trainer comprehensible, form of strength, speed and endurance are prerequisite to the fencer's ability to execute the fencing technique quickly, accurately and reliably, to perceive the opponent's behavior, make accurate decisions and set goals, feel anger or joy (left arrow to the top).

On the other hand, the action control occurs when the fencer sets goals, becomes motivated to achieve them, devises action programs, and chooses certain goals, ways, and means. Only then does the execution or conversion respectively, of the goal and situation appropriate movement programs (motor implementation) and the activation of the energy providing processes (right arrow to the bottom) occur.

The entirety of the performance and performance action is determined at the stimulus regulation level. The action, motion, and energy regulation is involved to a certain degree. In this context, what is important to the trainer is the knowledge that the stimulus level is superior to the action regulation level, in contrast the motion and energy level is inferior.

Competition failure or just one error can be caused by a decision that is not situation appropriate for an incorrect action (i.e. counter attack instead of parry-riposte), as well as

a tardy (missed time) or inaccurate movement in the sequence (too big; too small) or the target (missed target). Or the fencer just didn't really feeL like it.

2.1.2 DEMAND STRUCTURE AND TRAINING TASKS

These performance structure issues become important to the trainer, when practicing and training by goal formulation, determination of the necessary training content to reach the goal, training means and training methods are to be directed to where the fencer trains according to the demands, and that the targeted physical-organic movement-regulating, psychological-cognitive and emotional-volitional prerequisites and processes are developed and linked in a structurally ideal (according to the requirements) way. Regulation that meets the demands must be learned and continuously perfected and stabilized.

In a much-simplified version one can imagine that the human organism is a self-regulating system, in which strain that is caused by athletic training acts as a "disturbance of equilibrium". The organism attempts to balance these and tries to adapt. It is the trainers' job to optimize these adjustments to the strain. Development (improvement) always occurs when an athlete's existent performance requisites are not sufficient, and he can only complete the task or demand under personal exertion.

The trainer finds his basis for orientation in the demand structure.

Demand analyses are established during individual analytical steps that are executed with precision, applying methods of training science specific to high-performance sports.

1. The task analysis asks about the characteristics and the structure the fencer has to cope with during competition, and the conditions under which this occurs (first column of the structure plan in table 2.1-2). For instance, during the fencing fight the desire to win is most essential at the hierarchical upper level; at the lowest level the strain capacity is the constant alternating of forward and backwards movements, during which the fencer cannot fatigue.
2. Derived from this are
   - the behavior patterns, actions and movements essential to task execution, and
   - the performance requisites inventory essential to behavior, actions and movements (second column of the structure diagram).

(Self-assurance and organic-muscular load tolerance are prerequisites to the examples mentioned above.)

3. When tasks executed during training (third column) correspond to tasks performed during competition (first column), then performance rerequisites (second column) of the character and structure essential to task execution, are developed.
4. Various performance-monitoring methods are used (fourth column) to review the requirements for meeting demands (second column) and training effectiveness (third column). This is done via observation and tests for a differentiated assessment or via "complex performance diagnostics" of the overall performance.

This applies to the "construction" of requirement profiles for long-term training plans, which focus more on a versatile foundation, as well as the optimum preparation of fencers for international championships. This provides the trainer with a kind of "time table" for his responsible job.

Table 2.1.-2, on page 34, shows the relationship between the demands of fencing and the necessary performance requisites, the training exercises and the monitoring. The intent is to simplify the complexity and the complex correlations with the aid of such a "structure plan", so that a trainer can follow them anytime, and continue to advance and adapt them creatively to concrete prerequisites and conditions, and make them applicable.

Demands, performance requisites and training implementation have an inner connection that was linked, in the overview, to the assertions regarding complexity, the hierarchy of the regulation levels, and self-organization.

Significant to training are observations in the horizontal rows on the respective regulation level, as well as in the vertical columns in terms of the mutual interdependencies.

For the trainer, the proportionate weighting of the sub-systems involved in the complex fencing performance is only reasonable within these limits.

In Chapters 2.2 to 2.5 the analytical steps for the individual regulation levels and performance factors are addressed, based on this performance structure model. Performance factors are "lifted" as sections from the schematic performance structure model (fig. 2.1.-1) and broken down further.